

Review: "Method of Determination of the Shortest Path of Mobile Robotic Platform in the Conditions of Limited Resources"

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1. Introduction

The paper "Method of Determination of the Shortest Path of Mobile Robotic Platform in the Conditions of Limited Resources" by O. M. Pavliuk, M. O. Medykovskyy, M. V. Mishchuk and A. O. Zabolotna (Visnyk of Vinnytsia Polytechnic Institute, 2025, No. 1) addresses route planning for mobile robotic platforms in industrial environments. The objective is to determine the shortest route in dynamic surroundings while respecting resource constraints, primarily the residual battery charge and the permitted execution time.

2. Methodology

The method divides the route map into a uniform grid and builds a matrix of static obstacles and vectors of dynamic ones. Dynamic obstacles are classified as living, namely personnel requiring the highest safety priority, and inanimate, such as other platforms or pallets. Federated learning predicts the segmental voltage drop of each battery according to the individual characteristics of its platform, and the normalised forecast serves as a resource constraint during the search. Four classical algorithms, A*, D*, D*Lite and M*, are evaluated by route length, cells traversed, execution time and residual charge. The system is implemented in Python and validated on data from the Formica 1 platform.

3. Results

On a test map of 10×10 cells all algorithms produced identical paths except D*, which traversed 18 cells instead of 19 and preserved a marginally higher residual charge. D* also demonstrated the shortest execution time, whereas A* proved approximately twice slower. The analysis established that only M* incorporates the residual charge constraint directly, while A*, D* and D*Lite do not account for it at all. The authors recommend D* for dynamic environments and D*Lite or M* for stable ones.

4. Key Insights

Insight 1. Algorithm selection is governed not by asymptotic complexity but by the constraints the cost function is able to represent, since three of the four algorithms cannot express the battery limitation at all. This reframes the criteria applied in my own work, where algorithms were compared mainly by execution time.

Insight 2. The incremental principle underlying D*, whereby only the nodes affected by a change are recomputed instead of the entire route, constitutes a general design pattern. Repairing only the invalidated portion of a previous result applies well beyond navigation, and I intend to use it for repeated computation over changing data.

Insight 3. Federated learning is introduced for a practical rather than a conceptual reason: identical batteries on identical platforms degrade differently, so a single global model would misrepresent every individual unit. The stated preference for a local server over a cloud one shows that decentralisation serves fault tolerance as much as privacy.

5. Conclusion

The principal contribution is a hybrid method combining classical shortest-path algorithms with a neural forecast of battery voltage, which introduces an energy constraint into route planning. The validation, however, uses a single platform and a 10×10 grid, where the differences between algorithms amount to one cell and several microseconds. Extending it to full-scale maps and several simultaneous platforms, which the authors indicate as further work, would strengthen the conclusions.